

RBG-48363

10 CFR 50.73

June 5, 2025

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report, 50-458/2025-001-00, Human Performance Error
Results in Manual Reactor Trip

River Bend Station – Unit 1
NRC Docket Nos. 50-458
Renewed Facility Operating License No. NPF-47

Entergy Operations, Inc. (Entergy) submits the enclosed Licensee Event Report (LER), 50-458/2025-001-00 for River Bend Station, Unit 1 (RBS). The events reported herein are reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A), any event or condition that resulted in manual or automatic actuation of any of the systems listed in 10 CFR 50.73(a)(2)(iv)(B).

This letter contains no new regulatory commitments.

Should you have any questions, please contact me at 601-368-5756.

Respectfully,



Digitally signed by John Giddens
DN: cn=John Giddens, o=Entergy,
ou=Regulatory Compliance,
email=jgidden@entergy.com
Date: 2025.06.05 12:28:52 -05'00'

JMG/ahv

Enclosure: LER 50-458/2025-001-00

cc: NRC Region IV Regional Administrator - Region IV
NRC Senior Resident Inspector - River Bend Station
NRC Office of Nuclear Security and Incident Response

Enclosure

RBG-48363

Licensee Event Report
50-458/2025-001-00



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name River Bend Station, Unit 1	☒ 050	2. Docket Number 00458	3. Page 1 of 2
	☐ 052		

4. Title
Human Performance Error Results in Manual Reactor Trip

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
4	8	2025	2025	001	00	06	05	2025	N/A	☐ 052	Docket Number
									Facility Name	☐ 052	Docket Number
									N/A		

9. Operating Mode 1	10. Power Level 081
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)(i)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ Other (Specify here, in abstract, or in NRC 366A).

12. Licensee Contact for this LER

Licensee Contact Joseph Brown	Phone Number (Include Area Code) (225) 381-4693
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
A	NA	NA	NA	Y					

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

At 1607 CDT on April 8, 2025, with River Bend Station, Unit 1 operating in Mode 1 at 81% power, control room operators responded to alarms received for a total loss of normal service water. The control room operators entered AOP-0009, Loss of Normal Service Water, and subsequently inserted a manual reactor trip.

The direct cause of the loss of normal service water was the result of non-licensed operators failing to close SWP-MOV55A block valve, SWP-V3302, prior to post maintenance testing. Two causal factors were identified during the investigation: (1) Non-licensed operators did not appropriately apply human performance tools and verification practices when performing equipment protection tagout task; and, (2) Operations supervisor did not ensure adequate pre-job brief in accordance with EN-OP-102 and EN-HU-102 during briefing of equipment protection tagout installation – specifically, the risks involved if the work was performed incorrectly, stop-work criteria, and the key role of engagement by both performer and peer checker.

This event is reportable in accordance with 10 CFR 50.73 (a)(2)(iv)(A), any event or condition that resulted in manual or automatic actuation of any of the systems listed in 10 CFR 50.73 (a)(2)(iv)(B).

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
River Bend Station, Unit 1	☒ 050	00458	YEAR	SEQUENTIAL NUMBER	REV NO.
	☐ 052		2025	001	00

NARRATIVE**Event Description**

At 1607 CDT on April 8, 2025, with River Bend Station (RBS), Unit 1 operating in Mode 1 at 81% power, control room operators responded to alarms received for a total loss of normal service water [BI]. The control room operators entered AOP-0009, Loss of Normal Service Water and subsequently inserted a manual reactor trip.

The control room operators entered the appropriate post trip procedures and appropriately responded to the uncomplicated reactor trip.

This event was reported on April 8, 2025, at 1911 EDT in accordance with 10 CFR 50.72(b)(2)(iv)(B) for reactor protection system actuation and 10 CFR 50.72(b)(3)(iv)(A) due to an expected reactor water level 3 isolation signal (EN 57655).

Event Cause

The direct cause of a loss of normal service water was the result of non-licensed operators failing to close SWP-V3302 [BI:SHV], SWP-MOV55A block valve, prior to post maintenance testing. During post maintenance testing, maintenance personnel requested Operations to open SWP-MOV55A [BI:ISV], Standby Cooling Tower Division 1 Return Isolation Valve, as the valve opened all normal service water pumps [BI:P] received a low-pressure lockout condition due to an open flow path to the Standby Cooling Tower.

Two causal factors were identified during the investigation: (1) Non-licensed operators did not appropriately apply human performance tools and verification practices when performing equipment protection tagout task; and, (2) Operations supervisor did not ensure adequate pre-job brief in accordance with EN-OP-102, Protective and Caution Tagging, and EN-HU-102, Human Performance Traps and Tools, during briefing of equipment protection tagout installation – specifically, the risks involved if the work was performed incorrectly, stop-work criteria, and the key role of engagement by both performer and peer checker.

Safety Assessment

The actual consequence of this human performance error was a loss of normal service water and a subsequent manual plant trip. Standby Service Water initiated as expected and Normal Service Water was restored 37 minutes after loss. There were no other actual consequences to general safety of the public, nuclear safety, industrial safety and radiological safety for this event.

Corrective Actions**Completed:**

- Performance management actions were taken with the operators involved in the event.

In progress:

- Operations Shift Managers perform standards reset blitz.
- Create and implement verification challenge activity utilizing the simulator for non-licensed individuals.
- Operations Management perform standards reset with Senior Reactor Operators on utilizing quizzing during reverse briefs/pre-job briefs to ensure operations clearly understand the risk, probability of one, all instructions, and components to be manipulated. The discussion will include "I own the risk" and risk presented by operator performance.

Previous Occurrences

No previous similar events at RBS have occurred within the past 5 years in which a component misposition results in a reactor trip.